

Q1.

```
shekharchess8@practical-sat:~$ sudo apt update
Get:1 file:/etc/apt/mirrors/debian.list Mirrorlist [30 B]
Get:2 file:/etc/apt/mirrors/debian-security.list Mirrorlist [39 B]
Hit:7 https://packages.cloud.google.com/apt google-compute-engine-bookworm-stable InRelease
Hit:3 https://deb.debian.org/debian bookworm InRelease
Hit:8 https://packages.cloud.google.com/apt cloud-sdk-bookworm InRelease
Hit:4 https://deb.debian.org/debian bookworm-updates InRelease
Hit:5 https://deb.debian.org/debian bookworm-backports InRelease
Hit:9 https://packages.cloud.google.com/apt google-cloud-ops-agent-bookworm-2 InRelease
Hit:6 https://deb.debian.org/debian-security bookworm-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
3 packages can be upgraded. Run 'apt list --upgradable' to see them.
shekharchess8@practical-sat:~$ sudo apt install docker.io -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  binutils binutils-common binutils-x86-64-linux-gnu cgroupfs-mount containedr criu git git-man libbinutils libctf-nobfd0 libctf0 liberror-perl libgdm-compat4 libgprofng0 libintl-perl
  libintl1 libjansson4 libmodule-find-perl libnet11 libnetfilter1 libnl-3-200 libperl5.36 libproc-processstable-perl libprotobuf32 libsort-naturally-perl libterm-readkey-perl needrestart
  patch perl perl-modules-5.36 python3-protobuf runc tini
Suggested packages:
  binutils-doc containernetworking-plugins docker-doc aufs-tools brfs-progs debostrep line rootlesskit xfsprogs zfs-fuse | zfsutils-linux git-daemon-run | git-daemon-symlinks git-doc
  git-email git-gui gitk gitweb git-cvs git-mediawiki git-svn needrestart-session | libnotify-bin iucode-tool ed diffutils-doc perl-doc libterm-readline-gnu-perl | libterm-readline-perl-perl
  make libtap-harness-archive-perl
The following NEW packages will be installed:
  binutils binutils-common binutils-x86-64-linux-gnu cgroupfs-mount containedr criu docker.io git git-man libbinutils libctf-nobfd0 libctf0 liberror-perl libgdm-compat4 libgprofng0
  libintl-perl libintl1 libjansson4 libmodule-find-perl libnet11 libnetfilter1 libnl-3-200 libperl5.36 libproc-processstable-perl libprotobuf32 libsort-naturally-perl
  libterm-readkey-perl needrestart patch perl perl-modules-5.36 python3-protobuf runc tini
0 upgraded, 34 newly installed, 0 to remove and 3 not upgraded.
Need to get 91.5 MB of archives.
After this operation, 406 MB of additional disk space will be used.
Get:1 file:/etc/apt/mirrors/debian.list Mirrorlist [30 B]
Get:2 https://deb.debian.org/debian bookworm/main amd64 perl-modules-5.36 all 5.36.0-7+deb12u3 [2815 kB]
Get:3 https://deb.debian.org/debian bookworm/main amd64 libgdm-compat4 amd64 1:23-3 [48.2 kB]
Get:4 https://deb.debian.org/debian bookworm/main amd64 libperl5.36 amd64 5.36.0-7+deb12u3 [4196 kB]
Get:5 https://deb.debian.org/debian bookworm/main amd64 perl amd64 5.36.0-7+deb12u3 [239 kB]
```

```
shekharchess8@practical-sat:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-06-13 05:10:23 UTC; 2min 57s ago
   TriggeredBy: ● docker.socket
     Docs: https://docs.docker.com
    Main PID: 2605 (dockerd)
      Tasks: 8
     Memory: 28.9M
        CPU: 550ms
     CGroup: /system.slice/docker.service
             └─2605 /usr/sbin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Jun 13 05:10:22 practical-sat dockerd[2605]: time="2026-06-13T05:10:22.867328957Z" level=info msg="[core] Channel Connectivity change to CONNECTING" module=grpc
Jun 13 05:10:22 practical-sat dockerd[2605]: time="2026-06-13T05:10:22.868897506Z" level=info msg="[core] Subchannel Connectivity change to READY" module=grpc
Jun 13 05:10:22 practical-sat dockerd[2605]: time="2026-06-13T05:10:22.868933030Z" level=info msg="[core] Channel Connectivity change to READY" module=grpc
Jun 13 05:10:22 practical-sat dockerd[2605]: time="2026-06-13T05:10:22.907944850Z" level=info msg="Loading containers: start."
Jun 13 05:10:23 practical-sat dockerd[2605]: time="2026-06-13T05:10:23.246144746Z" level=info msg="Default bridge (docker0) is assigned with an IP address 172.17.0.0/16. Daemon option --bip c
Jun 13 05:10:23 practical-sat dockerd[2605]: time="2026-06-13T05:10:23.416414167Z" level=info msg="Loading containers: done."
Jun 13 05:10:23 practical-sat dockerd[2605]: time="2026-06-13T05:10:23.454340886Z" level=info msg="Docker daemon" commit=5d6db84 graphdriver(s)=overlay2 version=20.10.24+dfsg1
Jun 13 05:10:23 practical-sat dockerd[2605]: time="2026-06-13T05:10:23.454934039Z" level=info msg="Daemon has completed initialization"
Jun 13 05:10:23 practical-sat systemd[1]: Started docker.service - Docker Application Container Engine.
Jun 13 05:10:23 practical-sat dockerd[2605]: time="2026-06-13T05:10:23.514869984Z" level=info msg="API listen on /run/docker.sock"
lines 1-22/22 (END)
```

```
shekharchess8@practical-sat:~$ docker --version
Docker version 20.10.24+dfsg1, build 297e128
shekharchess8@practical-sat:~$
```

```
shekharchess8@practical-sat:~$ sudo systemctl start docker
shekharchess8@practical-sat:~$ sudo systemctl enable docker
Synchronizing state of docker.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable docker
shekharchess8@practical-sat:~$
```

```
shekharchess8@practical-sat:~$ sudo docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
72c03230f136: Pull complete
d4dcde3aeed: Pull complete
c5a7565de4cf: Pull complete
57b3fbf43092: Pull complete
80fa08b690ad: Pull complete
41c9f6f90940: Pull complete
6376488be516: Pull complete
Digest: sha256:608a100c71651bf5b773c89083b4a1ad7ef4b2bd05d7a7e552271e03123692ad
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
shekharchess8@practical-sat:~$
```

```
shekharchess8@practical-sat:~$ sudo docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
nginx                latest             936fef290c8f       2 days ago         161MB
shekharchess8@practical-sat:~$
```

```
shekharchess8@practical-sat:~$ docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
nginx          latest    936fef290c8f   2 days ago    161MB
shekharchess8@practical-sat:~$
```



Q2.

```
shekharchess8@practical-sat:~$ docker volume create studentdata
studentdata
shekharchess8@practical-sat:~$ docker volume ls
DRIVER      VOLUME NAME
local       my-volume
local       studentdata
shekharchess8@practical-sat:~$
```

```
shekharchess8@practical-sat:~$ docker run -it --name student-container -v studentdata:/data ubuntu bash
root@1e8d53833ed5:/#
```

```
shekharchess8@practical-sat:~$ docker run -it --name student-container -v studentdata:/data ubuntu bash
root@1e8d53833ed5:/# echo "Shekhar Chess - Roll No: YOUR_ROLL_NUMBER" > /data/info.txt
root@1e8d53833ed5:/# ls /data
info.txt
root@1e8d53833ed5:/#
```

```
root@1e8d53833ed5:/# echo "Shekhar singh - Roll no:24102A0050" > /data/info.txt
root@1e8d53833ed5:/# cat /data/info.txt
Shekhar singh - Roll no:24102A0050
root@1e8d53833ed5:/#
```

Q3.

```
shekharchess8@practical-sat:~$ docker build -t practical-exam:v1 .
Sending build context to Docker daemon 14.34kB
Step 1/2 : FROM ubuntu:latest
--> 30ba44506a6d
Step 2/2 : CMD echo "Cloud Computing Practical Exam Completed Successfully"
--> Running in 7a8a70bfd289
Removing intermediate container 7a8a70bfd289
--> 471758f7ee19
Successfully built 471758f7ee19
Successfully tagged practical-exam:v1
shekharchess8@practical-sat:~$
```

```
shekharchess8@practical-sat:~$ docker build -t practical-exam:v1 .
Sending build context to Docker daemon 14.34kB
Step 1/2 : FROM ubuntu:latest
--> 30ba44506a6d
Step 2/2 : CMD echo "Cloud Computing Practical Exam Completed Successfully"
--> Running in 7a8a70bfd289
Removing intermediate container 7a8a70bfd289
--> 471758f7ee19
Successfully built 471758f7ee19
Successfully tagged practical-exam:v1
shekharchess8@practical-sat:~$ docker images
REPOSITORY          TAG             IMAGE ID         CREATED          SIZE
practical-exam      v1             471758f7ee19    34 seconds ago  100MB
nginx               latest         936fef290c8f    2 days ago      161MB
ubuntu              latest         30ba44506a6d    7 weeks ago     100MB
shekharchess8@practical-sat:~$ docker run practical-exam:v1
Cloud Computing Practical Exam Completed Successfully
shekharchess8@practical-sat:~$
```